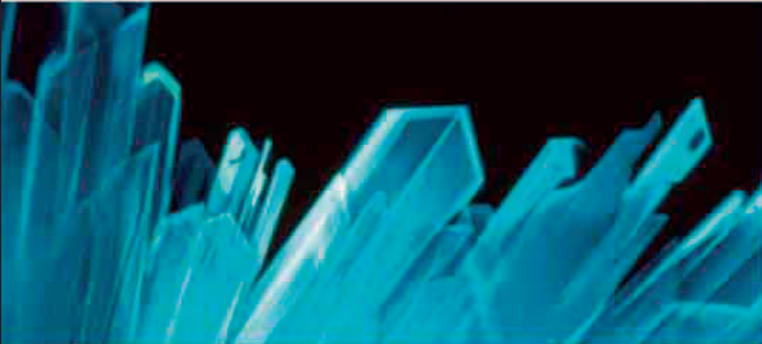




Sodium sulfate crystals, known as Glauber's salt up to the late 18th century, are mostly sourced from natural minerals.

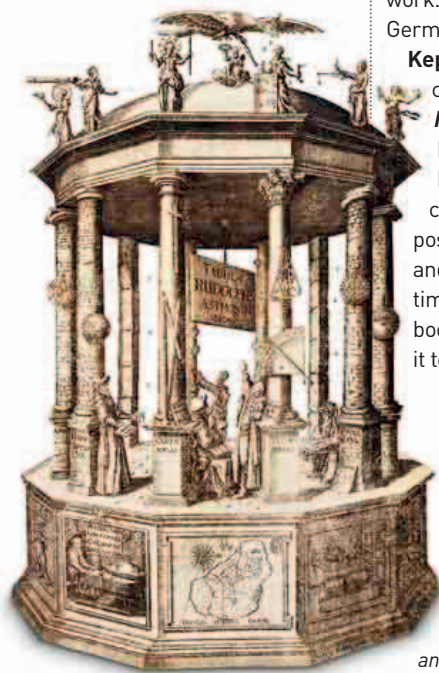
IN 1625, YOUNG DUTCH-GERMAN CHEMIST JOHANN GLAUBER

(1604–70) recovered from a stomach bug after drinking from a spring. The following year, he succeeded in crystallizing *sal mirabile* (miraculous salt) from the spring's water. This became known as **Glauber's salt** and is in fact **sodium sulfate**, which has laxative properties. For nearly 300 years physicians would use it as a purgative.

In 1626, while traveling through icy London, English philosopher **Francis Bacon**—champion of the idea that theories must be built

up from empirical evidence—acted on an impulse. He wanted to see if he could preserve meat by stuffing a chicken carcass with snow. While the experiment was a success, Bacon contracted **pneumonia** and never recovered.

In 1627, the most accurate **catalog of astronomical measurements** was published since Nicolaus Copernicus had suggested that the Sun was at the center of the Solar System (see 1543). Much of the data had been collected by Danish astronomer **Tycho Brahe**, but he died before he could publish the work. It fell to his collaborator, German astronomer **Johannes Kepler**, to complete the catalog, which was named **Rudolphine Tables** after the Holy Roman Emperor Rudolph II. This work contains data on the positions of nearly 1,500 stars, and the planets known at the time. Kepler financed the book's printing and dedicated it to Brahe.

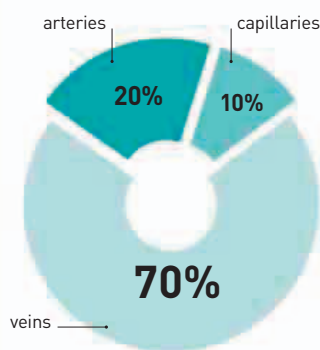


Frontispiece of Rudolphine Tables

This depicts an imaginary monument celebrating the achievement of generations of astronomers, including Hipparchus, Ptolemy, Nicolaus Copernicus, and Tycho Brahe.

“THE **HEART OF ANIMALS** IS THE FOUNDATION OF THEIR LIFE... THE **SUN OF THEIR MICROCOSM**, UPON WHICH ALL GROWTH DEPENDS, FROM WHICH **ALL POWER PROCEEDS.**”

William Harvey, English physician, from *An Anatomical Essay*, 1628



Distribution of blood

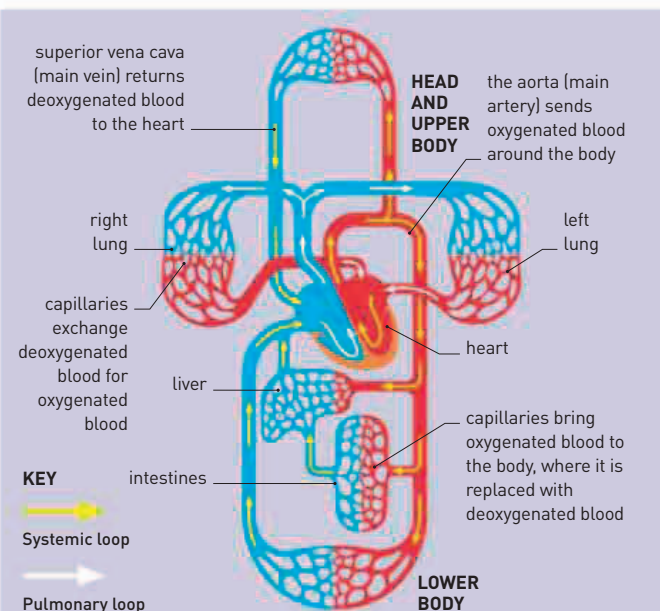
Most blood in the systemic circulation is carried in the veins. The blood in these thin-walled vessels is at a very low pressure, so it effectively pools there.

IN 1628, ENGLISH PHYSICIAN WILLIAM HARVEY published his most celebrated work: *An Anatomical Essay Concerning the Movement of the Heart and Blood in Animals*. Harvey was a great believer in the need for science to progress by experimentation, and had closely studied the **blood systems of animals**. In the previous century, Italian physician Matteo Colombo had demonstrated that the heart worked as a pump, and not by suction, as the Ancient Greeks had thought. But the traditional view persisted—originating with Ancient Greek surgeon, and philosopher Galen—that blood was continuously made in the liver. After assessing the pumping effect of the heart,

Harvey doubted this was true. His experiments had shown that so much blood was pumped by the heart that continuous production was improbable. Instead, he deduced that the volume of blood is fixed and this is continuously circulated in the body. High-pressure blood,

coming from the heart, is distributed around the body through arteries, while low pressure blood returns through veins. He also theorized about a specific **circulation for the lungs**.

In 1629, Italian inventor **Giovanni Branca** published



CIRCULATION OF BLOOD

A double blood circulation system facilitates the exchange of oxygen and carbon dioxide, and ensures maximum pressure of blood in the lungs and around the body. Blood that has been oxygenated in the lungs (red) is pumped around the body by the left side of the heart. Deoxygenated blood from body tissues (blue) is pumped back to the lungs by the heart's right side.

1625 Dutch-German chemist Johann Glauber discovers a **laxative salt**

1626 Santorio Santorio is the first to measure human body temperature with a **thermometer**

1627 Adriaan van den Spiegel describes the development of the **human fetus**

1628 William Harvey publishes *An Anatomical Essay Concerning the Movement of the Heart and Blood in Animals*

1629 John Parkinson publishes *Park-in-Sun's Terrestrial Paradise*, said to be the **first gardening book**

1625 Christoph Scheiner verifies Johannes Kepler's prediction that an **inverted image** forms on the retina

October 31, 1626 Joseph of Arimatari provides evidence that chick **embryos** are "preformed" from fertilization

1627 Kepler completes *Rudolphine Tables*

1629 Giovanni Branca publishes a collection of inventions that includes an early **steam engine**